Solution:

Here the number of rows and columns are equal.

∴ The given assignment problem is balanced.

Now let us find the solution.

Step 1: Select a smallest element in each row and subtract this from all the elements in its row.

	I	II	III	IV
A	0	2	9	1
B	0	5	2	3
C	1	3	2	0
D	0	7	3	1

Look for atleast one zero in each row and each column.Otherwise go to step 2.

Step 2: Select the smallest element in each column and subtract this from all the elements in its column.

	I	II	III	IV
A	0	0	7	1
B	0	3	0	3
C	1	1	0	0
D	0	5	1	1

Since each row and column contains atleast one zero, assignments can be made.

Step 3 (Assignment):

Examine the rows with exactly one zero. First three rows contain more than one zero. Go to row *D*. There is exactly one zero. Mark that zero by □(i.e) job *D* is assigned to machine I. Mark other zeros in its column by ×.

	I	II	III	IV
A	~~0~~	0	7	1
B	~~0~~	3	0	3
C	1	1	0	0
D	□	5	1	2

Step 4: Now examine the columns with exactly one zero. Already there is an assignment in column I. Go to the column II. There is exactly one zero. Mark that zero by □. Mark other zeros in its row by ×.

	I	II	III	IV
A	~~0~~	□	7	1
B	~~0~~	3	0	3
C	1	1	0	0
D	□	5	1	2

Column III contains more than one zero. Therefore proceed to Column IV, there is exactly one zero. Mark that zero by □. Mark other zeros in its row by ×.

	I	II	III	IV
A	~~0~~	□	7	1
B	~~0~~	3	0	3
C	1	1	~~0~~	□
D	□	5	1	2

Step 5: Again examine the rows. Row B contains exactly one zero. Mark that zero by □.

	I	II	III	IV
A	~~0~~	□	7	1
B	~~0~~	3	□	3
C	1	1	~~0~~	□
D	□	5	1	2

Thus all the four assignments have been made. The optimal assignment schedule and total cost is

Job	Machine	cost
A	II	12
B	III	7
C	IV	11
D	I	8
Total cost		38

XII Std - Business Maths & Stat EM Chapter 10.indd 253 2/29/2020 11:45:20 AM